# Q3(d) 2023 Midterm: Hicksian Demand Explanation

ECO 310

## 1 The Question

Harry's preferences are $u(N, C) = C^2 + 2CN$ (where $\alpha = 2$).

**Part (d)(ii):** Netflix and cable are free, but Harry has limited time. How much time $T$ does he need to achieve utility $u^*$?

## 2 Why This Is a Hicksian/Expenditure Problem

This question asks: *"What's the minimum amount of time (expenditure) needed to achieve a target utility level $u^*$?"*

This is **exactly** the Hicksian demand problem from your study guide:

**Hicksian Problem:**

$$\min_x p \cdot x \quad \text{subject to} \quad u(x) \geq \bar{u}$$

In this specific question:

- The "expenditure" is **time** $T = C + N$ (not money)
- The "prices" are both 1 (each hour of Netflix or Cable costs 1 hour of time)
- The constraint is achieving utility level $u^*$

## 3 Step-by-Step Solution

### 3.1 Step 1: Set Up the Problem

We want to minimize time expenditure:

$$\min_{C,N} T = C + N$$

subject to achieving the target utility:

$$u(N, C) = C^2 + 2CN \geq u^*$$

### 3.2 Step 2: Find the Optimal Allocation (Tangency Condition)

At the optimum, the marginal rate of substitution equals the price ratio:

$$\frac{MU_C}{MU_N} = \frac{p_C}{p_N}$$

Calculate the marginal utilities:

$$MU_C = \frac{\partial u}{\partial C} = 2C + 2N$$
$$MU_N = \frac{\partial u}{\partial N} = 2C$$

Therefore:

$$\frac{MU_C}{MU_N} = \frac{2C + 2N}{2C} = \frac{C + N}{C} = 1 + \frac{N}{C}$$

Since both "prices" are 1 (one hour each):

$$1 + \frac{N}{C} = \frac{p_C}{p_N} = \frac{1}{1} = 1$$

This gives us:

$$\frac{N}{C} = 0 \quad \Rightarrow \quad \boxed{N = 0}$$

### 3.3 Step 3: Find the Hicksian Demand

Now we use the utility constraint. With $N = 0$:

$$u^* = C^2 + 2C(0) = C^2$$

Solving for $C$:

$$\boxed{C = \sqrt{u^*}}$$

This is the **Hicksian demand** for cable: $h_C(p_C = 1, p_N = 1, u^*) = \sqrt{u^*}$
And the Hicksian demand for Netflix is: $h_N(p_C = 1, p_N = 1, u^*) = 0$

### 3.4 Step 4: Find the Expenditure Function

The total time needed (expenditure function) is:

$$\boxed{T = e(p_C = 1, p_N = 1, u^*) = C + N = \sqrt{u^*} + 0 = \sqrt{u^*} \text{ hours}}$$

## 4 Key Insights

### 4.1 Why Does Harry Choose Zero Netflix?

When both goods have the same "price" (1 hour each), Harry compares their marginal utilities:

- $MU_C = 2C + 2N$ (marginal utility of cable)
- $MU_N = 2C$ (marginal utility of Netflix)

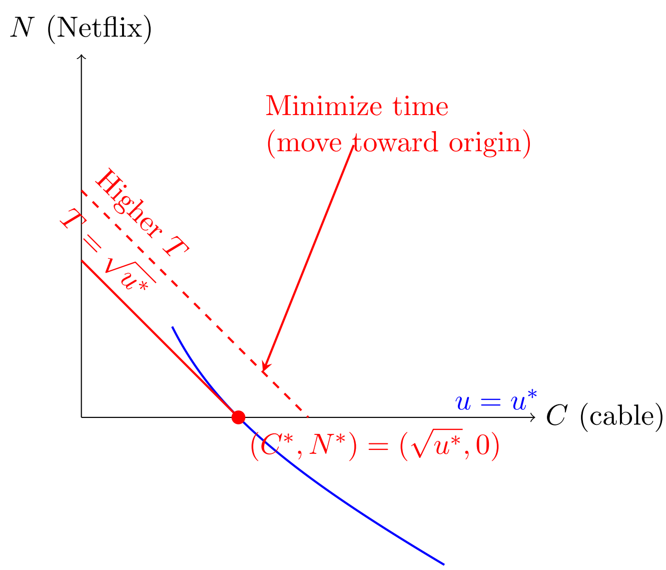
Since $MU_C > MU_N$ whenever $N > 0$, cable gives more "bang per buck" (or bang per hour). So Harry specializes completely in cable.

### 4.2 Connection to Marshallian Demand

Compare this to part (c), which found the **Marshallian demand** (income $m$ is fixed):

| **Type** | **What's Fixed?** | **Question** |
|---|---|---|
| Marshallian (part c) | Income/budget $m$ | "What should I buy with my budget?" |
| Hicksian (part d) | Utility level $u^*$ | "What's the cheapest way to get utility $u^*$?" |

### 4.3 Graphical Interpretation




The Hicksian problem finds the point on the indifference curve $u = u^*$ that minimizes time expenditure $T = C + N$. The optimal point is $(\sqrt{u^*}, 0)$ on the horizontal axis.

## 5 Summary

Part (d)(ii) is a **Hicksian/expenditure minimization problem**:

1. **Goal:** Minimize time expenditure to achieve utility $u^*$
2. **Method:** Use tangency condition $\frac{MU_C}{MU_N} = \frac{p_C}{p_N}$ with constraint $u = u^*$
3. **Result:** Hicksian demands are $h_C = \sqrt{u^*}$ and $h_N = 0$
4. **Expenditure function:** $e(1, 1, u^*) = \sqrt{u^*}$ hours

This contrasts with part (c), which solved the Marshallian problem: maximize utility given a fixed budget $m$.